CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EC)

April/May 2012

EC304 Power Electronics

Date: 03.05.2012, Thursday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1	(a)	Define (i)Forward recovery time (ii)Reverse recovery time (iii) Softness factor in power diode.	[03]
	(b)	Draw and explain the SCR in detail also explain SCR V-I characteristics with required terms.	[04]
Q - 2	(a)	Enumerate the methods by which thyristor is turned on. Explain any two.	[04]
	(b)	Draw a snubber circuit for SCR. How does it provides dv/dt protection?	[04]
	(c)	Explain the working of single phase unidirectional controller.	[06]
		OR	
Q - 2	(a)	An SCR with latching current 50 mA is connetced in series with resistance of 20Ω and inductnace of 0.5H.The duration of the firing pulse is 50μSec. Will the thyristor get fired.	[04]
	(b)	Discuss one quadrant chopper fed dc drive for separately excited dc motor.	[04]
	(c)	Explain the operation of 1-Φ dual converter with waveforms.	[06]
Q - 3	(a)	Explain the basic parallel inverter with waveforms.	[06]
	(b)	Draw the circuit and output waveforms for 3-Φ Full converter with highly inductive load for firing angle (α) of $\pi/3$. Derive the equations for output DC and RMS voltages for $\alpha \leq \pi/3$.	[08]
		OR	
Q - 3	(a)	The forward voltage drop of power diode is V_D =1.2 V at I_D = 300mA. Assuming that n=2 and V_T =25.8 mV, Find the saturation current I_s.	[03]
	(b)	Why it is necessary to connect thyristors in parallel? Explain the problems associated with it. How it can be overcome?	[03]
	(c)	Explain the operation of three phase to single phase cycloconverter.	[08]

SECTION – II

Q - 4(a) Justify that for high frequency operation MOSFET is used instead of BJT as switching device. [03]

(b) List two major drawbacks of square wave inverters. How it can be overcome? [04]

Q - 5 (a) Draw the circuit of buck-boost regulator and explain its working with equivalent circuit and waveforms. Obtain equation for ΔI and ΔVc . [08]

(b) Explain the operating modes of DC motor. [06]

OR

Q - 5 (a) Explain the operation of 3-phase bridge inverter operating in 120^0 mode. Draw appropriate waveforms. [08]

(b) Draw and explain Half wave controlled rectifier with resistive load. Draw the waveforms. Derive the equations for average load voltage and RMS load voltage. [06]

Q - 6 (a) In a step up chopper applied voltage is 200V and output voltage is 600V. If the conducting time of thyristor is 200μSec, Compute [08]

i) chopper frequency

ii) Find output voltage when the pulse width is half for constant frequency of operation.

Derive the necessary equations.

(b) Explain the construction and characteristics of IGBT. [06]

OR

Q - 6 (a) Explain the operation of 1-phase bridge inverter with suitable waveforms. Derive necessary equations. [08]

(b) Write note on Chopper classification. [06]

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